Name: _____

TOPIC TEST 1

Chapter 12: Applications of derivatives

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 8 multiple-choice questions (8 marks)
- Part B: 10 short answer questions (50 marks)
- Part C: 4 analysis questions (22 marks)

Part A

8 multiple-choice questions
1 mark each: 8 marks
Circle the correct answer.

1 The rate at which a waterfall is flowing over a cliff is $R = 4t + 3t^2$ m^{3}s^{-1}. Find the amount of water flowing after a minute if the amount of water is 10 970 m^3 after 20 seconds.

A 223 220 m^3

B 8800 m^3

C 225 370 m^3

D 226 250 m^3

E 109 700 m^3

[1 mark]

2 $g(x) = x^2 + 5x - 3$. The average rate of change from $x = 0$ to $x = 2$ is:

A 9

B 4

C 3

D 5

E 7

[1 mark]

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3 When the function $f(x)$ has a minimum:

A $f'(x) = 0$ and increases

B $f'(x) > 0$ and increases

C $f'(x) < 0$ and increases

D $f'(x) = 0$ and decreases

E $f'(x) > 0$ and decreases.

[1 mark]

4 The values of a function and its derivative for selected points are given in the table. All values for which the derivative is zero are shown. The function has no asymptotes.

t	-3	-1	1	2	5
$f(t)$		4		2	
$f'(t)$	-4	0	-2	0	3

The function has:

A a maximum at $(2, 2)$

B a point of inflection at $(-1, 4)$

C a minimum at $(-1, 0)$

D a minimum at $(2, 0)$

E a point of inflection at $(2, 2)$.

[1 mark]

5 The function $f(x)$ has a derivative that can be factorised to $f'(x) = (3 - x)(x + 4)$. $f(x)$ has:

A a minimum at $x = 3$ and a maximum at $x = -4$

B a maximum at $x = 3$ and a minimum at $x = -4$

C a point of inflection at $x = -4$ and a maximum at $x = 3$

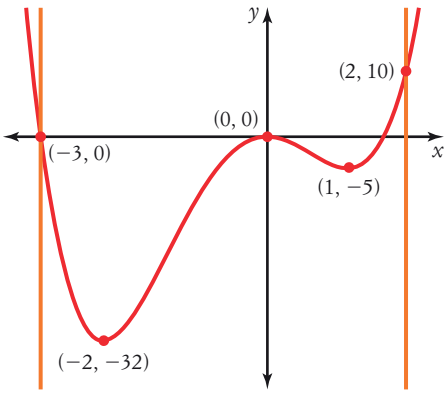
D a maximum at $x = 3$ and a point of inflection at $x = -4$

E points of inflection at $x = -4$ and $x = 3$.

[1 mark]

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The diagram below shows the graph of a function, $f(x)$. Use the graph to answer questions **6** to **8**, for the domain $-3 \leq x \leq 2$.



6 A local maximum for this function can be found at:

A $(-2, -32)$

B $(-3, 0)$

C $(1, -5)$

D $(2, 10)$

E $(0, 0)$

[1 mark]

7 The greatest value on the specified domain can be found at:

A $(-2, -32)$

B $(-3, 0)$

C $(1, -5)$

D $(2, 10)$

E $(0, 0)$

[1 mark]

8 The least value on the specified domain can be found at:

A $(-2, -32)$

B $(-3, 0)$

C $(1, -5)$

D $(2, 10)$

E $(0, 0)$

[1 mark]

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Part B

10 short answer questions
50 marks
Show your working where appropriate.

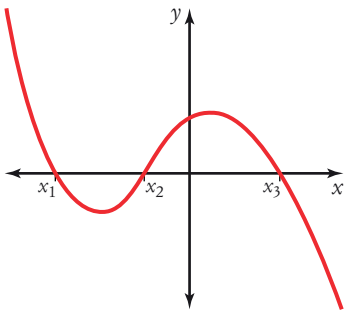
1 If $f(x) = 3x^5 - 2x^4 + x^3 - 2$, find:

a $f(-1)$

b $f'(-1)$

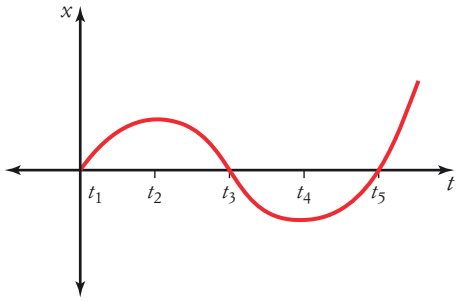
[5 marks]

2 Sketch a primitive function for this graph.



[5 marks]

3 The graph below shows the displacement of a particle.



a When is the particle at the origin? _____

b When is it at rest? _____

c When is it travelling at its greatest speed? _____

d Sketch the graph of its:

i velocity

ii acceleration.

[9 marks]

4 The height of a ball thrown in the air is given by the function $h = 20t - 5t^2$, where h is in metres and t is the time in seconds after the ball has been thrown.

a Find the ball's height after 1 s.

b What is the maximum height of the ball?

c How long is the ball in the air for?

[6 marks]

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5 A curve has $\frac{dy}{dx} = 6x^2 + 12x - 5$. If the curve passes through the point $(2, -3)$, find the equation of the curve.

[3 marks]

6 A curve passes through the point $(0, -1)$ and the gradient at any point is given by $(x + 3)(x - 5)$. Find the equation of the curve.

[4 marks]

7 The acceleration of a particle is given by $6t - 12$ ms^{-2}. If the particle is initially at rest 2 m to the left of the origin, find its displacement after 5 seconds.

[5 marks]

8 Find all x values for which the curve $y = 3x^2 - 18x + 4$ is decreasing.

[3 marks]

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9 Show that the function $f(x) = 2(5x - 3)^3$ has a point of inflection at (0.6, 0).

[4 marks]

10 Find the greatest value of the curve $y = x^3 + 3x^2 - 24x - 1$ in the domain $-5 \leq x \leq 6$.

[6 marks]

Part C

4 analysis questions
22 marks
Show your working where appropriate.

1 The rate of change of V with respect to t is given by $\frac{dV}{dt} = (2t - 1)^2$. If $V = 5$ when $t = \frac{1}{2}$, find V when $t = 3$.

[5 marks]

2 A particle moves so that its displacement after t seconds is $x = 4t^2 - 5t^3$ metres. Find:

a its initial displacement, velocity and acceleration

b when the particle is at the origin

c its maximum displacement.

[7 marks]

3 Find the stationary points on the curve $y = x^3 + 6x^2 + 9x - 11$ and determine their nature.

[5 marks]

4 The volume of air in a cubic room on a submarine is given by the formula

$V = -4x^3 + 27x^2 - 24x + 2$, where x is the side length of the room in metres. Find the dimensions of the room that will give the maximum volume of air.

[5 marks]
Total marks: 80

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Answers

Part A [1 mark each]

1 Volume (v) = antiderivative of $R = 4t^2 + t^3 + c$

When $t = 20$, $v = 10\,970 = 4 \times 20^2 + 20^3 + c$; so $c = -5030$

When $t = 60$, $v = 4 \times 60^2 + 60^3 - 5030 = 225\,370$.

$\therefore$ **C** **[1 mark]**

2 Average rate of change $= \dfrac{g(2) - g(0)}{2 - 0} = \dfrac{11 - (-3)}{2} = 7$

$\therefore$ **E** **[1 mark]**

3 For a minimum, the gradient (derivative) goes from negative to zero to positive.

$\therefore$ **A** **[1 mark]**

4 When $t < -1, f'(t) < 0$, when $t = -1, f'(t) = 0$ and when $t > -1, f'(t) < 0$. Point of inflection at $t = -1$.

$\therefore$ **B** **[1 mark]**

5 Stationary points exist where $f'(x) = 0$, i.e. $x = 3$ or $x = -4$.

When $x < -4, f'(x) < 0$, when $x = -4, f'(x) = 0$ and when $x > -4, f'(x) > 0$. Minimum at -4.

When $x < 3, f'(x) > 0$, when $x = 0, f'(x) = 0$ and when $x > 3, f'(x) < 0$. Maximum at 3.

$\therefore$ **B** **[1 mark]**

6 There is only 1 local maximum. It occurs at (0, 0).

$\therefore$ **E** **[1 mark]**

7 The greatest value of the function over the domain $-3 \leq x \leq 2$ is 10. It occurs at (2, 10).

$\therefore$ **D** **[1 mark]**

8 The least value of the function over the domain $-3 \leq x \leq 2$ is -32. It occurs at $(-2, -32)$.

$\therefore$ **A** **[1 mark]**

Part B

1 a $f(-1) = 3 \times (-1)^5 - 2 \times (-1)^4 + (-1)^3 - 2$ **[1 mark]**

$= -8$ **[1 mark]**

b $f'(x) = 15x^4 - 8x^3 + 3x^2$ **[1 mark]**

$f'(-1) = 15 \times (-1)^4 - 8 \times (-1)^3 + 3 \times (-1)^2$ **[1 mark]**

$= 26$ **[1 mark]**

[5 marks]

2 Draw a set of axes and mark in points x_1, x_2 and x_3 in the same relative positions as they are on the graph of the function.

The primitive function will have stationary points at x_1, x_2 and x_3.

The primitive function will have:

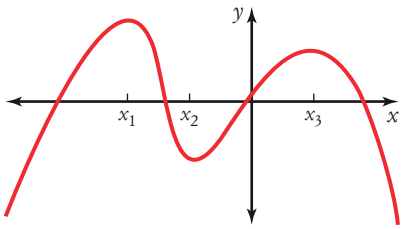
- a positive slope from the left up to x_1 **[1 mark]**
- a negative slope between x_1 and x_2 **[1 mark]**

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- a positive slope between x_2 and x_3 **[1 mark]**
- a negative slope from x_3 and beyond. **[1 mark]**

Draw the graph. **[1 mark]**

[5 marks]



3 a The particle is at the origin when the graph crosses the t-axis. This occurs at t_1, t_3 and t_5. **[1 mark]**

b The particle is at rest when the velocity is 0. This occurs at the stationary points at t_2 and t_4. **[1 mark]**

c The greatest speed corresponds to the steepest slope of the graph. This occurs at t_5. **[1 mark]**

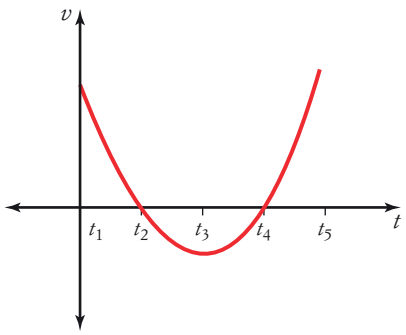
d Velocity is the derivative of displacement.

Draw up a set of axes and mark in the points at t_1 to t_5 on the t-axis. **[1 mark]**

The velocity graph will:

- have zeros at t_2 and t_4
- be positive from t_1 to t_2 and from t_4 to the right
- be negative between t_2 and t_4. **[1 mark (for all features)]**

Draw the graph showing these features. **[1 mark]**



e Acceleration is the derivative of velocity.

Draw up a set of axes and mark in the points at t_1 to t_5 on the t-axis. **[1 mark]**

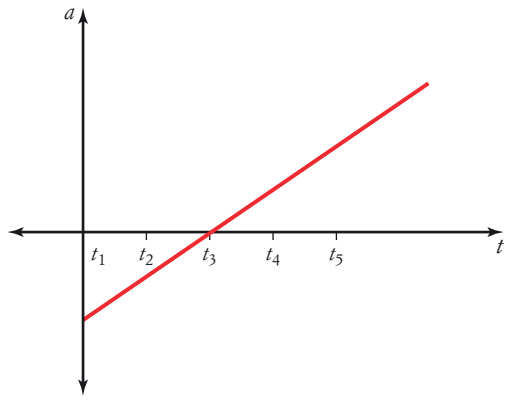
The acceleration graph will have:

- a zero at t_3
- be negative from t_1 to t_3
- be positive from t_3 to the right. **[1 mark (for all features)]**

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Draw the graph showing these features. **[1 mark]**

[9 marks]



4 a $h = 20 \times 1 - 5 \times 1^2 = 15$ m **[1 mark]**

b Maximum height is reached when the velocity $= 0$

$h' = v = 20 - 10t$ **[1 mark]**

Let $v = 0$; $20 - 10t = 0$, so $t = 2$.

$h = 20 \times 2 - 5 \times 2^2 = 20$ m **[1 mark]**

c Ball is in the air while $h > 0$.

Let $h = 0$.

$20t - 5t^2 = 0$ **[1 mark]**

$5t(4 - t) = 0$

$\therefore t = 0$ or 4 **[1 mark]**

The ball is on the ground at 0 s and 4 s.

The ball is in the air for 4 s. **[1 mark]**

[6 marks]

5 $y = 2x^3 + 6x^2 - 5x + c$ **[1 mark]**

When $x = 2$, $y = -3$

$-3 = 2 \times 2^3 + 6 \times 2^2 - 5 \times 2 + c$ **[1 mark]**

$\therefore c = -33$

$y = 2x^3 + 6x^2 - 5x - 33$ **[1 mark]**

[3 marks]

6 $\dfrac{dy}{dx} = (x + 3)(x - 5) = x^2 - 2x - 15$ **[1 mark]**

$y =$ antiderivative of the gradient

$= \dfrac{x^3}{3} - x^2 - 15x + c$ **[1 mark]**

When $x = 0, y = -1$

$-1 = 0 - 0 - 0 + c$

$\therefore c = -1$ **[1 mark]**

$y = \frac{x^3}{3} - x^2 - 15x - 1$ **[1 mark]**

[4 marks]

7 Velocity is the antiderivative of acceleration.

$v = \frac{6t^2}{2} - 12t + c = 3t^2 - 12t + c$ **[1 mark]**

When $t = 0, v = 0$, so $c = 0$ and $v = 3t^2 - 12t$ **[1 mark]**

Displacement is the antiderivative of velocity

$s = \frac{3t^3}{3} - \frac{12t^2}{2} + d = t^3 - 6t^2 + d$ **[1 mark]**

When $t = 0, s = -2$, so

$-2 = 0 - 0 + d$

$\therefore d = -2$

and $s = t^3 - 6t^2 - 2$ **[1 mark]**

When $t = 5, s = 5^3 - 6 \times 5^2 - 2 = -27$m **[1 mark]**

[5 marks]

8 $y' = 6x - 18$ **[1 mark]**

Let $y' = 0$.

$6x - 18 = 0$

$\therefore \quad x = 3$ **[1 mark]**

Stationary point at $x = 3$.

When $x < 3, y' < 0$, so y is decreasing when $x < 3$. **[1 mark]**

[3 marks]

9 $f(x) = 250x^3 - 450x^2 + 270x - 54$ **[1 mark]**

$f'(x) = 750x^2 - 900x + 270x$

$= 30(5x - 3)^2$ **[1 mark]**

Let $f'(x) = 0$.

$30(5x - 3)^2 = 0$

$\therefore x = 0.6$ **[1 mark]**

$f'(x) > 0$ for all values of x except $x = 0$, and $f'(x) = 0$ at $x = 0.6$, so there is a point of inflection at $x = 0.6$. **[1 mark]**

[4 marks]

10 $y' = 3x^2 + 6x - 24 = 3(x - 2)(x + 4)$ **[1 mark]**

Let $y' = 0$.

$3(x - 2)(x + 4) = 0$

$\therefore x = -4$ or 2 **[1 mark]**

When $x < -4$, $y' > 0$; $x = -4$, $y' = 0$; $x > -4$ (but < 2), $y' < 0$. i.e. a local maximum. **[1 mark]**

When $x = -4$, $y = 79$, so there is a local maximum at $(-4, 79)$.

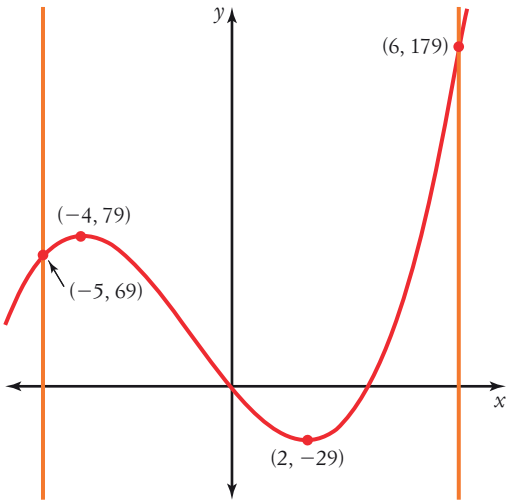
When $x < 2$ (but > -4), $y' < 0$; $x = 2$, $y' = 0$; $x > 2$, $y' > 0$, i.e. a local minimum. **[1 mark]**

When $x = 2$, $y = -29$, so there is a local minimum at $(2, -29)$.

When $x = -5$, $y = (-5)^3 + 3 \times (-5)^2 - 24 \times (-5) - 1 = 69$

When $x = 6$, $y = 6^3 + 3 \times 6^2 - 24 \times 6 - 1 = 179$ **[1 mark (for both boundary values)]**

Draw a sketch.



The global maximum value in the range $-5 \leq x \leq 6$ is 179. **[1 mark]**

[6 marks]

Part C

1 $\dfrac{dV}{dt} = 4t^2 - 4t + 1$ **[1 mark]**

$V = \dfrac{4t^3}{3} - \dfrac{4t^2}{2} + t + c = \dfrac{4t^3}{3} - 2t^2 + t + c$ **[1 mark]**

When $t = \dfrac{1}{2}$, $V = 5$.

$\dfrac{4\left(\frac{1}{2}\right)^3}{3} - 2\left(\frac{1}{2}\right)^2 + \frac{1}{2} + c = 5$ **[1 mark]**

$\Rightarrow c = 4\dfrac{5}{6}$

$\therefore V = \dfrac{4t^3}{3} - 2t^2 + t + 4\dfrac{5}{6}$ **[1 mark]**

Let $t = 3$.

$\frac{4 \times 3^3}{3} - 2 \times 3^2 + 3 + 4\frac{5}{6} = 25\frac{5}{6}$

When $t = 3$, $V = 25\frac{5}{6}$ **[1 mark]**

[5 marks]

2 a When $t = 0$, $x = 4 \times 0^2 - 5 \times 0^3 = 0$ m **[1 mark]**

$v = \frac{dx}{dt} = 8t - 15t^2$

When $t = 0$, $v = 8 \times 0 - 15 \times 0^2 = 0$ m/s **[1 mark]**

$a = \frac{dv}{dt} = 8 - 30t$

When $t = 0$, $a = 8 - 15 \times 0 = 8$ m/s^2 **[1 mark]**

b Let $x = 0$.

$4t^2 - 5t^3 = 0$

$t^2(4 - 5t) = 0$

$\therefore t = 0$ s or 0.8 s **[1 mark]**

c Let $\frac{dx}{dt} = 0$.

$8t - 15t^2 = 0$

$t(8 - 15t) = 0$

$\therefore t = 0$ s or $\frac{8}{15}$ s **[1 mark]**

When $t < 0$, $\frac{dx}{dt} < 0$; when $t = 0$, $\frac{dx}{dt} = 0$; and when $t > 0$ (but $< \frac{8}{15}$), $\frac{dx}{dt} > 0$.

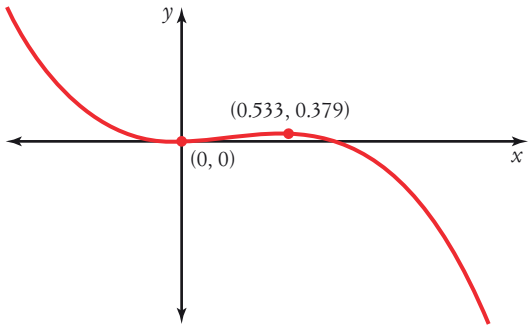
Minimum at 0.

When $t < \frac{8}{15}$ (but > 0), $\frac{8}{15} > 0$, when $t = \frac{8}{15}$, $\frac{dx}{dt} = 0$ and when $t > \frac{8}{15}$, $\frac{dx}{dt} < 0$.

Maximum at $\frac{8}{15}$. **[1 mark]**

When $t = \frac{8}{15}$, $x = 4 \times \left(\frac{8}{15}\right)^2 - 5 \times \left(\frac{8}{15}\right)^3 = \frac{256}{675} \approx 0.379$ m

Draw a sketch.



The maximum displacement is about 0.38 m. **[1 mark]**

[7 marks]

3 $y' = 3x^2 + 12x + 9$ **[1 mark]**

Let $y' = 0$.

$3x^2 + 12x + 9 = 0$

$3(x + 1)(x + 3) = 0$

$\therefore x = -1$ or -3 **[1 mark]**

When $x < -3$, $y' > 0$; when $x = -3$, $y' = 0$; and when $x > -3$ (but < -1), $y' < 0$. Maximum at -3. **[1 mark]**

When $x < -1$ (but > -3), $y' < 0$; when $x = -1$, $y' = 0$; and when $x > -1$, $y' > 0$. Minimum at -1. **[1 mark]**

When $x = -3$, $y = (-3)^3 + 6 \times (-3)^2 + 9 \times (-3) - 11 = -11$

When $x = -1$, $y = (-1)^3 + 6 \times (-1)^2 + 9 \times (-1) - 11 = -15$

[1 mark]

The function has a local maximum at $(-3, -11)$, and a local minimum at $(-1, -15)$. **[5 marks]**

4 $V' = -12x^2 + 54x - 24$ **[1 mark]**

Let $V' = 0$.

$-12x^2 + 54x - 24 = 0$

$-6(x - 4)(2x - 1) = 0$

$\therefore x = 4$ or $\frac{1}{2}$ **[1 mark]**

When $x < \frac{1}{2}$, $V' < 0$; when $x = \frac{1}{2}$, $V' = 0$; and when $x > \frac{1}{2}$ (but < 4), $V' > 0$.

Minimum at $\frac{1}{2}$. **[1 mark]**

When $x < 4$ (but $> \frac{1}{2}$), $V' > 0$; when $x = 4$, $V' = 0$; and when $x > 4$, $V' < 0$.

Maximum at 4. **[1 mark]**

The room is cubic, so its dimensions are 4 m $\times$ 4 m $\times$ 4 m **[1 mark]**

[5 marks]

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